XORFLOW: Supplementary Evaluation Appendix

Anonymous Authors

I. Supplementary evaluation scope

This supplement reports support writes, anchor-lifecycle accounting, structural controls, codec bytes, shared-memory/SCALE-Sim scheduling, held-out Ramulator2 validation, RTL throughput, and physical-design measurements. Producer and consumer anchor sources are accounted for explicitly, every producer recovery precedes target encoding, and one common scheduler is applied to every compatible checkpoint.

II. Inter-layer structure

Figure S1 directly tests the paper's empirical premise. Exact-count independent masks preserve each layer's number of active positions, yet raise the median event density by $8.38\times$ relative to real adjacent pairs. Row permutation produces median Jaccard 0.447 and forces fallback for essentially every record. Nonadjacent layer $\ell+2$ remains more correlated than random but is consistently weaker than the adjacent pair. The comparison isolates learned positional correspondence from density and within-layer row regularity.

III. Codec decomposition

Independent majority-prototype coding reduces physical bytes by 5.4% relative to optimized BEICSR. Fixed-anchor XOR reaches 10.5%, generic XOR–gap coding 10.4%, forced delta 19.6%, and complete online selection 12.7%. Temporal coding is therefore the largest increment beyond independent spatial coding. Forced delta is smaller in aggregate but loses local protection and the fixed-partition pair-write bound. The complete XORFLOW rows use the same serializer, anchor accounting, and execution model as the primary evaluation. Ablation conclusions are restricted to serialized and physical bytes.

IV. Consumer accounting and dependency audit

The 16-KiB consumer store retains 422 of 72,604 delta anchors (0.581%); 72,182 recoveries reread 142.67 MiB and consume 1.320 million decode cycles. The producer path independently rereads the same total, giving 285.34 MiB and 2.641 million decode cycles across both lifecycles. At 256 KiB, 1 MiB, and 4 MiB, consumer hits rise to 7.10%, 27.94%, and 80.43%, while rereads fall to 133.46, 103.92, and 28.61 MiB.

The per-record audit reports zero producer-readiness violations across 73,652 targets and classifies every one of the 72,604 delta targets. The common scheduler covers all ten eligible checkpoints plus a 16-layer Arxiv extension and Chameleon control. Its independent max-plus recurrence agrees exactly with every baseline and proposal schedule.

V. External memory and combination validation

The memory model uses one persistent eight-channel resource for producer reads, consumer reads, and writebacks; a 32-entry request queue is separate from eight active channel slots. One trace calibrates the timing scale, and an independent held-out trace is predicted at 4,064,919 cycles and completes in 4,083,428 Ramulator2 cycles, an absolute error of 0.453%. Ramulator2 uses HBM2, RoBaRaCoCh mapping, FR-FCFS, open rows, refresh, and finite read/write queues. DRAMsim3 independently verifies complete service under a second mapping.

VI. Producer and consumer resource sensitivity

The producer surface spans input depth 1/2/4/8, one/two/four workers, and outstanding capacity 2/4/8/16. Worker parallelism dominates the cycle model: one, two, and four workers reach roughly 69, 138, and 271–278 input bits/cycle. Deeper input queues increase staging but improve four-worker throughput only modestly. On the consumer, bank conflicts rather than lane parsing limit the eight-bank case; 32 banks reduce conflicts from 138,496.5 to 3,992 and sustain 1,822.4 encoded bits/cycle.

VII. Robustness and operating regime

The slice-width result is stable: the complete support representation changes little from width 64 to 256 on the representative Arxiv, Reddit, and Yelp traces, indicating that the gain is not tuned to a single slice width. Cache sensitivity is workload dependent because feature misses can either expose or dilute support traffic. Exact O1 edge ordering retains 13.65% Arxiv and 33.30% Reddit traffic reduction. The bandwidth panel is a byte-roofline projection and is used only to explain which layers are memory exposed.

VIII. Quality and implementation

Every primary paired FP8/FP16 quality point lies close to the FP32 result; the maximum absolute difference is 0.053 percentage points. Table S6 reports all ten compatible primary pairs. The execution model reports $1.220\times$ for 16-layer Arxiv and $0.994\times$ for the Chameleon control. CACTI results quantify the bank-count trade-off of the support cache, while routed and synthesized logic results are reported separately below.

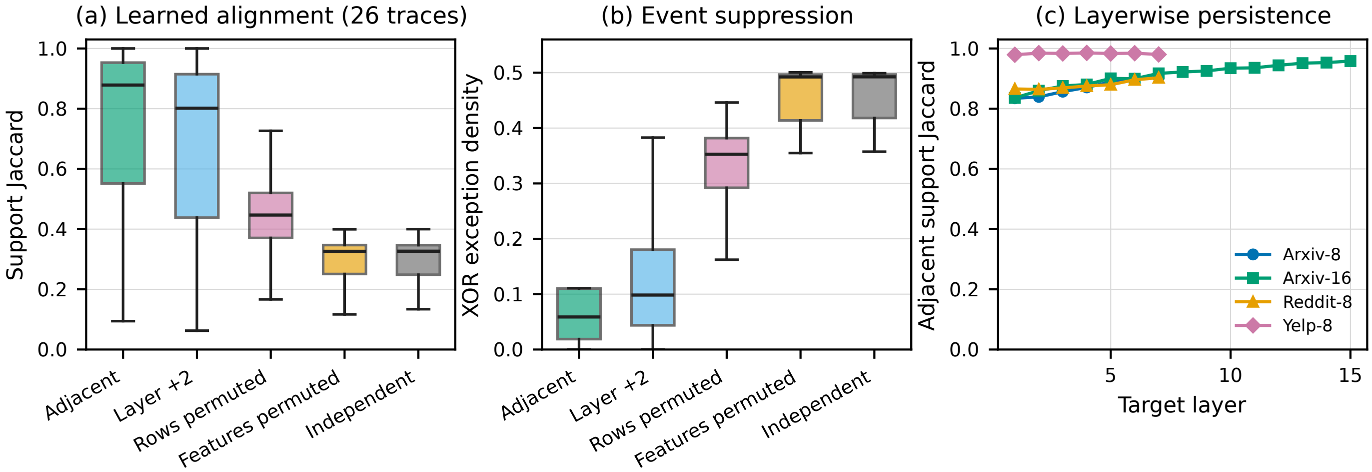

Fig. S1. Complete support-persistence characterization. (a–b) Distributions across 26 traced configurations. Real adjacent supports have median Jaccard 0.879 and median XOR exception density 0.0588; exact-count independent controls reach 0.327 and 0.4925. Row and feature permutations destroy most of the advantage while preserving important marginal statistics. (c) Representative layerwise traces show that persistence remains high through residual depth. Source: `A1_layerwise_persistence.csv` and `A2_learned_structure_controls.csv`.

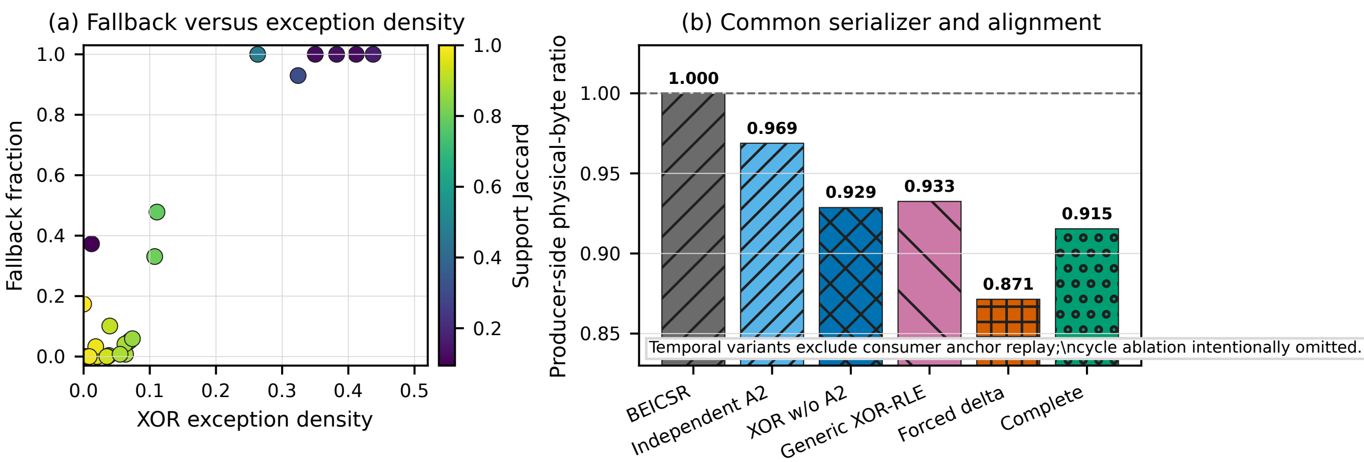

Fig. S2. Common-accounting codec decomposition. (a) Physical-byte reduction uses the same serializer, padding, unchanged traffic, and consumer anchor bytes for every variant. (b) All variants use common byte accounting; timing comparisons are reported for the baseline and complete XORFLOW design. Source: `results/ablation_bytes_only.csv`.

IX. Detailed Derivations and Proofs

A. Hamming-optimal cohort prototype

For cohort rows $x_1, \ldots, x_g \in \{0,1\}^C$ and prototype $P \in \{0,1\}^C$, the residual event count is

$$E(P) = \sum_{r=1}^{g} \|x_r \oplus P\|_0 \tag{1}$$

$$= \sum_{r=1}^{g} \sum_{j=1}^{C} \mathbf{1}[x_r[j] \neq P[j]] \tag{2}$$

$$= \sum_{j=1}^{C} \sum_{r=1}^{g} \mathbf{1}[x_r[j] \neq P[j]]. \tag{3}$$

Define $n_j = \sum_{r=1}^{g} x_r[j]$. For a fixed column j, choosing $P[j] = 0$ incurs

$$E_j(0) = \sum_{r=1}^{g} \mathbf{1}[x_r[j] = 1] = n_j, \tag{4}$$

whereas choosing $P[j] = 1$ incurs

$$E_j(1) = \sum_{r=1}^{g} \mathbf{1}[x_r[j] = 0] = g - n_j. \tag{5}$$

Because columns are independent in the objective,

$$\min_{P \in \{0,1\}^C} E(P) = \sum_{j=1}^{C} \min_{p \in \{0,1\}} E_j(p) \tag{6}$$

$$= \sum_{j=1}^{C} \min(n_j, g - n_j). \tag{7}$$

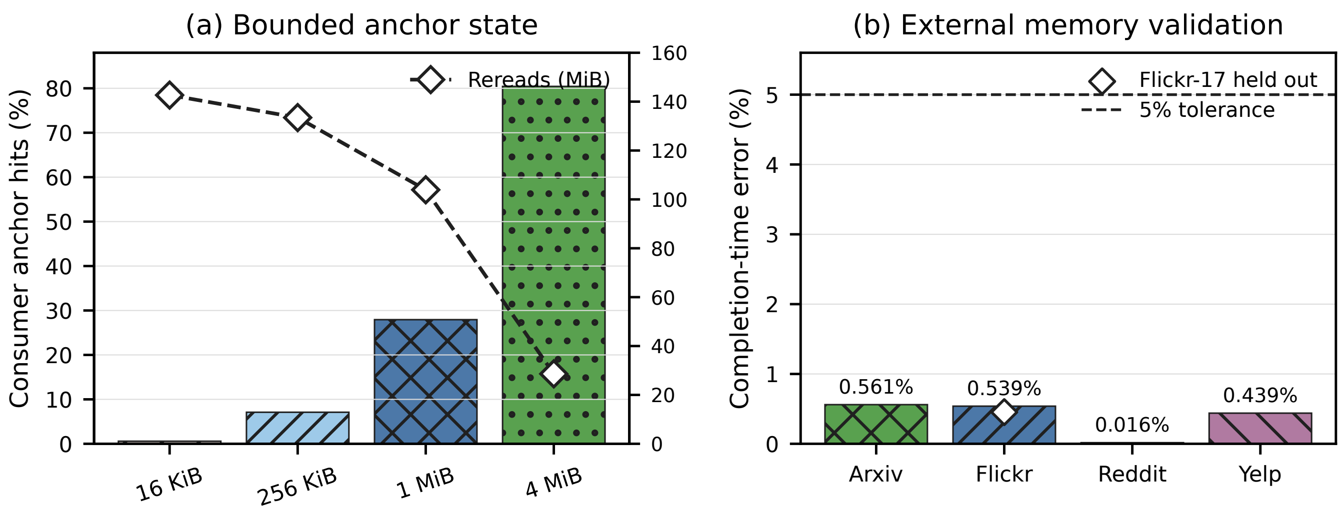

Fig. S3. Lifecycle and external-memory validation. (a) Consumer hit rate rises with anchor capacity while exact compressed rereads fall. (b) Complete retained-stream predictions for Arxiv, Flickr, Reddit, and Yelp remain within 0.57% of Ramulator2; the Flickr–17 diamond is an independent held-out absolute check. Source: `results/anchor_lifecycle_summary.csv` and `results/complete_external_memory_validation.csv`.

TABLE S1
FINAL RESULTS FOR EVERY COMPATIBLE EIGHT-LAYER CHECKPOINT. SUPPORT AND PHYSICAL TRAFFIC ARE EXACT; CYCLES ARE MODELED AGGREGATION–COMBINATION-SUBSYSTEM COMPLETION TIMES FROM THE COMMON SHARED-MEMORY/SCALE-SIM SCHEDULE.

Trace	Support reduction	Physical reduction	BEICSR cycles	XORFLOW cycles	Speedup
Flickr–7	56.75%	5.88%	46,213,044	43,803,355	1.055×
Flickr–17	58.14%	0.22%	46,271,863	46,455,417	0.996×
Flickr–27	55.33%	-1.93%	46,256,780	47,454,167	0.975×
Arxiv–7	21.56%	9.88%	118,054,997	107,599,087	1.097×
Arxiv–17	19.49%	9.69%	117,773,363	107,576,129	1.095×
Arxiv–27	18.68%	11.54%	117,893,069	105,498,473	1.117×
Reddit–7	27.12%	21.81%	5,949,006,307	4,650,397,803	1.279×
Reddit–17	33.85%	9.96%	6,073,413,354	5,467,747,091	1.111×
Reddit–27	34.34%	7.54%	6,073,613,917	5,615,099,894	1.082×
Yelp–7	45.24%	4.80%	740,915,129	708,134,575	1.046×

Thus an optimizer is

$$P^*[j] = \begin{cases} 1, & n_j > g/2, \\ 0, & n_j \le g/2, \end{cases} \tag{8}$$

with zero selected on ties. Since $\min(n_j, g-n_j) \le g/2$ for every j,

$$E(P^*) = \sum_{j=1}^{C} \min(n_j, g-n_j) \le \sum_{j=1}^{C} \frac{g}{2} = \frac{gC}{2}. \tag{9}$$

The result is nontrivial for hardware because the globally optimal binary prototype is obtained by C independent population counts and comparisons; no iterative or combinatorial search is required.

B. Fixed-partition non-expansion bound

Let q denote fixed tile boundaries, slice width, headers, and alignment. The anchor selector evaluates a set containing the independent baseline record, hence

$$B_A^{\mathrm{XORFLOW}}(A_p) = \min_{F\in\{\mathrm{BEICSR,A0,A2}\}} B_F^{(q)}(A_p) \tag{10}$$

$$\le B_{\mathrm{BEICSR}}^{(q)}(A_p). \tag{11}$$

The target selector compares the delta and independent records, hence

$$B_T^{\mathrm{XORFLOW}}(A_p, T_p) = \min\big(B_\Delta(A_p \oplus T_p), \tag{12}$$

$$B_{\mathrm{BEICSR}}^{(q)}(T_p)\big) \tag{13}$$

$$\le B_{\mathrm{BEICSR}}^{(q)}(T_p). \tag{14}$$

Adding these inequalities gives

$$B_{\mathrm{pair}}^{\mathrm{XORFLOW}}(A_p, T_p) = B_A^{\mathrm{XORFLOW}}(A_p) + B_T^{\mathrm{XORFLOW}}(A_p, T_p) \tag{15}$$

$$\le B_{\mathrm{BEICSR}}^{(q)}(A_p) + B_{\mathrm{BEICSR}}^{(q)}(T_p). \tag{16}$$

The inequality holds record by record and therefore also after summing over any trace. It bounds serialized support writes under the common partition q; producer/consumer rereads

TABLE S2

DEPTH, WIDTH, AND OPERATOR TRANSFER. PRIMARY/COMMON-SCHEDULER ROWS AND CONSUMER-COMPLETE SENSITIVITY ROWS ARE IDENTIFIED SEPARATELY; BORDERLINE AND INVALID-QUALITY MODELS ARE DIAGNOSTICS ONLY. ALL SPEEDUPS ARE MODELED AGGREGATION–COMBINATION-SUBSYSTEM RESULTS VERSUS OPTIMIZED BEICSR.

Dataset	Model variation	Setting	Quality/status	Speedup	Accounting scope	Role
Arxiv	DeepRes depth	D8	valid	1.103×	consumer-complete	depth anchor
Arxiv	DeepRes depth	D16	valid	**1.217**×	consumer-complete	valid deep point
Arxiv	DeepRes depth	D24	67.21%, borderline	**1.252**×	consumer-complete extension	diagnostic
Arxiv	DeepRes depth	D32	66.70%, borderline	**1.314**×	consumer-complete extension	diagnostic
Reddit	DeepRes depth	D8	95.34%, valid	**1.272**×	consumer-complete	depth anchor
Reddit	DeepRes depth	D12	95.46%, valid	**1.302**×	consumer-complete extension	valid deep point
Reddit	DeepRes depth	D16	95.60%, valid	**1.324**×	consumer-complete extension	valid deep point
Flickr	DeepRes depth	D8	47.23%, valid	1.059×	consumer-complete	depth anchor
Flickr	DeepRes depth	D16	47.34%, valid	0.981×	consumer-complete extension	negative control
Yelp	DeepRes depth	D8	43.40%, borderline	1.045×	consumer-complete	depth anchor
Yelp	DeepRes depth	D12	45.90%, valid	1.098×	consumer-complete extension	valid deep point
Yelp	DeepRes depth	D16	37.99%, invalid	1.050×	consumer-complete extension	diagnostic
Arxiv	DeepRes width	F64	valid	0.991×	finite-queue sensitivity	narrow boundary
Arxiv	DeepRes width	F128	valid	1.113×	finite-queue sensitivity	reference
Arxiv	DeepRes width	F256	valid	1.181×	finite-queue sensitivity	width scaling
Arxiv	Residual GIN	D8/F128	valid	1.156×	finite-queue sensitivity	operator transfer
Arxiv	Residual GraphSAGE	D8/F128	valid	**1.248**×	finite-queue sensitivity	operator transfer
Arxiv	Non-residual GIN	D8/F128	valid	1.003×	finite-queue sensitivity	fallback control

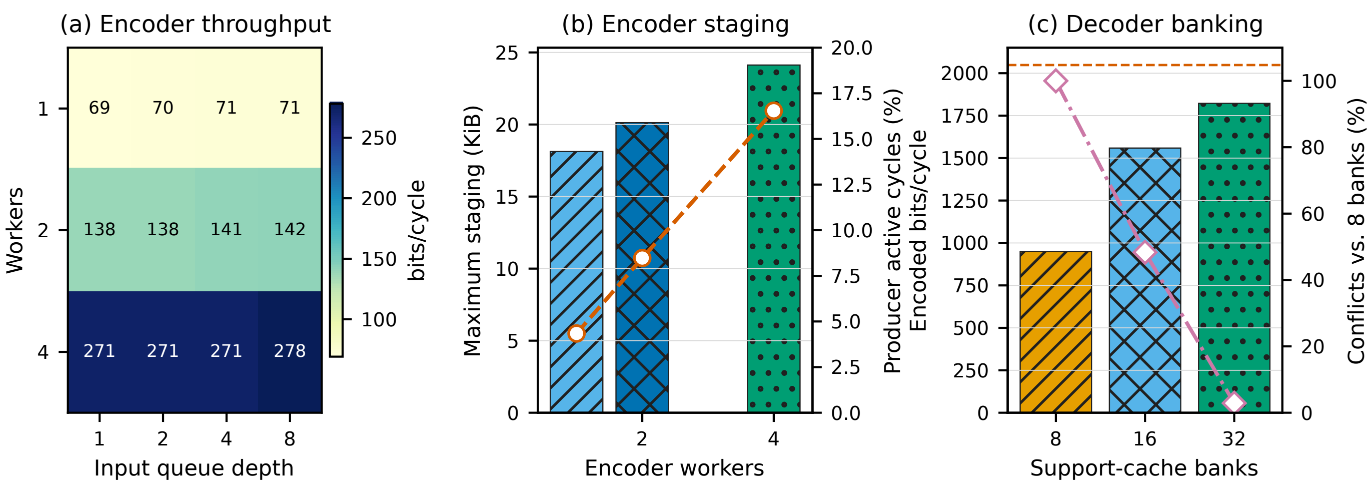

Fig. S4. Finite producer queues and banked reconstruction. (a) Encoder throughput scales primarily with worker count; increasing outstanding-format capacity has negligible throughput effect. (b) Four workers reach 278.1 input bits/cycle with at most 25 KiB of staging in the evaluated surface. Producer stall fraction is measured against an always-ready source and therefore quantifies offered-load saturation. (c) Increasing support-cache banks from 8 to 32 raises median decoder throughput from 949.6 to 1,822.4 encoded bits/cycle, raises lane utilization from 46.4% to 89.0%, and reduces conflicts by 97.1%. Source: `A7_encoder_queue_surface.csv` and `A8_decoder_banking.csv`.

TABLE S3

CODEC DECOMPOSITION UNDER COMMON SERIALIZER, PADDING, UNCHANGED-TRAFFIC, AND ANCHOR-ACCOUNTING RULES. NON-COMPLETE VARIANTS SUPPORT BYTE ATTRIBUTION ONLY.

Representation	Physical-byte reduction vs. BEICSR
Independent A2	5.4%
Fixed-anchor XOR	10.5%
Generic XOR–Gap	10.4%
Forced delta	**19.6%**
Complete online XORFLOW	12.7%

and other traffic components remain explicit in the system evaluation.

C. *Exact reconstruction*

For each target pair, $D_p = A_p \oplus T_p$. Using associativity, commutativity, and self-inversion of XOR,

$$A_p \oplus D_p = A_p \oplus (A_p \oplus T_p) \tag{17}$$
$$= (A_p \oplus A_p) \oplus T_p \tag{18}$$
$$= 0 \oplus T_p \tag{19}$$
$$= T_p. \tag{20}$$

The equality holds independently for every bit, so reconstruction is exact for the complete support bitmap.

X. REPRODUCIBILITY MATERIALS

The accompanying materials include support and physical-byte tables, absolute cycles for all ten eligible checkpoints,

TABLE S4

IMPLEMENTATION AND VALIDATION SUMMARY. ROUTED LOGIC, SRAM ESTIMATES, STREAM THROUGHPUT, AND EXTERNAL TIMING ARE REPORTED AS DISTINCT EVIDENCE CLASSES.

Component/check	Configuration	Result	Tool/evidence	Interpretation
Producer tile front end	32×64 support region	1,055 mapped cells	Yosys	Majority + XOR discovery mapped
Producer streams	Dense/ID/Gap8, randomized stalls	24/24 exact matches	RTL/software co-sim	Byte-exact finite interface
Decoder model	32 lanes, 32 banks	1,822.4 encoded bit/cycle	Bank-aware replay	89.0% lane utilization
Integrated consumer	Eight lanes, Nangate45	+0.565 ns at 1 GHz; 0 DRC	OpenROAD/ORFS	Routed timing closure
Consumer die/cells	Eight-lane cluster	0.0682 mm^2 die; 0.001795 mm^2 cells	OpenROAD/ORFS	Logic area reported separately
Support SRAM	16 KiB, 45 nm	0.0549 mm^2; 0.374 ns; 0.00891 nJ/read	CACTI	Fits 1-GHz access target
Memory timing	Flickr–17 held out	0.453% completion error	Ramulator2	Below predeclared 5% tolerance
Schedule cross-check	20 primary schedules	0.0 recurrence error	Independent max-plus model	Exact deterministic agreement
Support reconstruction	All evaluated streams	Bit-exact	Software/RTL tests	No approximation introduced

TABLE S5

HELD-OUT ABSOLUTE MEMORY VALIDATION AND VERSIONED SCALE-SIM COMBINATION SHAPES. THE INTERNAL TIMING SCALE USES FLICKR–7 ONLY; FLICKR–17 IS THE HELD-OUT CASE.

Memory case	Raw internal	Predicted	Ramulator2	Error	Role
Flickr–7	2,381,344	4,004,721	4,004,721	0.000%	Calibration
Flickr–17	2,417,140	4,064,919	4,083,428	0.453%	Held out

M	K	N	SCALE-Sim cycles	Utilization
5	128	128	1,583	5.05%
34	128	128	2,047	26.58%
47	128	128	2,255	33.35%
127	128	128	3,535	57.48%
128	128	128	3,551	57.67%
101	64	64	779	51.86%
128	64	64	887	57.72%

TABLE S6

PAIRED NUMERICAL QUALITY FOR ALL COMPATIBLE PRIMARY CHECKPOINTS. Δ IS FP8/FP16 MINUS FP32 IN PERCENTAGE POINTS; THE ARITHMETIC CONTRACT IS SHARED BY THE BASELINE AND XORFLOW.

Trace	FP32	FP8/FP16	Δ (pp)	Trace	FP32	FP8/FP16	Δ (pp)
Flickr–7	0.472460	0.472281	-0.018	Arxiv–7	0.686727	0.687036	+0.031
Flickr–17	0.470354	0.470174	-0.018	Arxiv–17	0.682756	0.682777	+0.002
Flickr–27	0.472550	0.472774	+0.022	Arxiv–27	0.687118	0.686583	-0.053
Reddit–7	0.953557	0.953360	-0.020	Reddit–17	0.947166	0.946969	-0.020
Reddit–27	0.947148	0.947094	-0.005	Yelp–7	0.433947	0.433952	+0.001

scheduler source, recurrence and dependency audits, SCALE-Sim shape data, Ramulator2 outputs, anchor accounting, negative controls, PPA, and CACTI data.

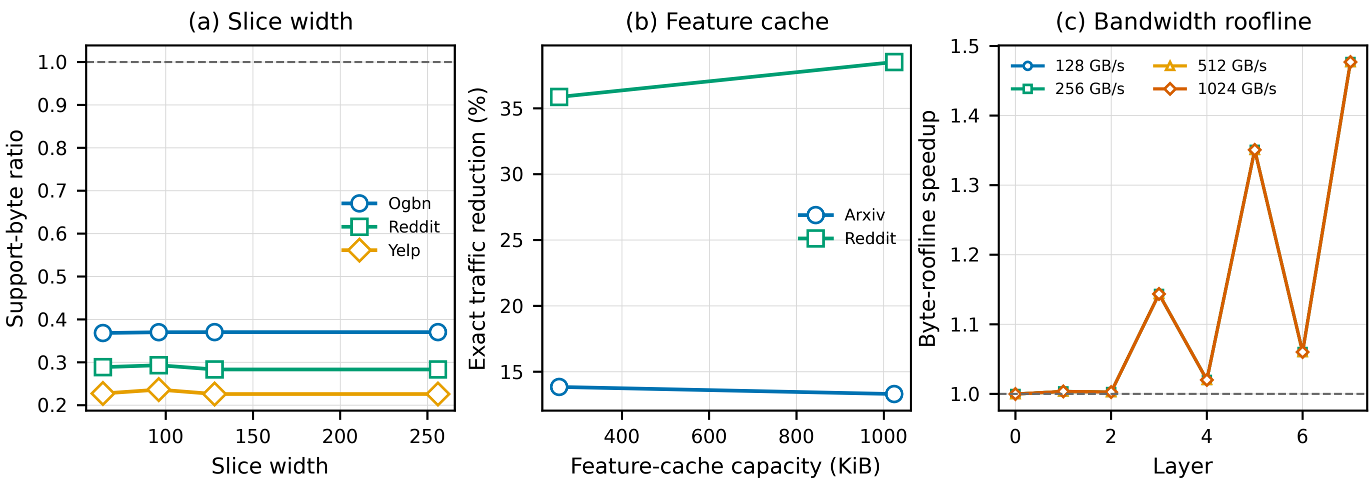

Fig. S5. Representation and memory sensitivity. (a) Complete support-byte ratios are stable across 64–256-feature slices for representative primary traces. (b) Increasing feature-cache capacity from 256 KiB to 1 MiB changes exact traffic reduction from 13.85% to 13.31% on Arxiv and from 35.87% to 38.52% on Reddit. (c) Byte-roofline projections identify layers whose traffic reduction is exposed under memory-bound execution; they identify layers whose traffic reduction is exposed under memory-bound execution. Source: `A9_slice_width.csv`, `A9_feature_cache_exact_traffic.csv`, and `A10_bandwidth_roofline_projection.csv`.